Bagus Budi Laksono

Lead ETL / Data Engineer · Legacy ETL Modernization (Ab Initio → Python / PySpark on Cloud)

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PROFESSIONAL SUMMARY

Lead ETL / Data Engineer with 7+ years of enterprise-scale data integration in telecommunications. Deep Ab Initio expertise (graph development, batch & continuous flow, Control Center, EME) combined with hands-on modernization of legacy ETL into reliable, config-driven **Python / PySpark pipelines running on a cloud data platform**. I own the full migration path: reverse-engineering legacy metadata, field-level mapping, re-implementation, **record-level parity validation** against legacy output, and cloud operation with idempotent, observable jobs. I also ship full-stack products (Next.js, TypeScript, Solidity).

CORE SKILLS

Legacy ETL	Ab Initio GDE (reformat, join, sort, rollup, partition, gather), DML/XFR, plans & psets, Control Center, EME check-in/out & dependency analysis
Modern data	Python, PySpark / Spark SQL, Pandas, PyArrow, Snowflake, Hive / HDFS, cloud object storage, SFTP / Kafka ingestion
Practices	Legacy-to-cloud migration, parity & reconciliation testing, data quality & quarantine, idempotent loads, lineage / DAG analysis, code generation, Shell / KSH, Git
Product & web	Next.js, React, TypeScript, Tailwind CSS, Solidity / Hardhat, ethers.js, Google Apps Script, PHP

PROFESSIONAL EXPERIENCE

XL Axiata Tbk — Lead ETL Developer

Feb 2019 – Present

Legacy ETL modernization: Ab Initio → Python / PySpark on cloud (current)

- Lead the migration of the legacy mediation framework (collect → parse → distribute) from Ab Initio to Python / PySpark jobs running on a cloud data platform.
- Assessed an inventory of 3,300+ legacy jobs, scoped 1,300+ for migration and planned delivery in waves; flagged retirement candidates for customer sign-off.
- Designed an engine-as-library architecture: shared collect / parse / distribute engines, self-contained per-feed rule modules and one unified orchestrator with dry-run and debug modes.
- Built code generators that produced 4,500+ stage runners and 3,300+ runner docs directly from legacy parameter sets.
- Achieved 100% record-level parity against legacy output on production feeds (e.g. 130K+ records, 39 fields, 0 mismatches), root-causing residual gaps down to defects in the legacy output itself.
- Delivered distribution to Snowflake and Hive, ingestion via SFTP / file transfer, and cloud object storage integration.

Ab Initio engineering & platform

- Led development of enterprise-scale Ab Initio ETL solutions and settlement platforms; built and maintained complex graphs for billing, settlement, reporting and reconciliation.
- Built a Python metadata reverse-engineering toolkit parsing .mp, .plan, .pset, .xfr and .dml into JSON DAGs (13-step pipeline), with interactive DAG visualizations for audit and impact analysis.
- Executed validation and mapping across ~200 sandbox environments, including graph / sub-graph relationship mapping and parameter extraction.
- Supported Control Center deployment, scheduling and operational validation; managed EME code (check-in/check-out, dependency analysis).
- Drove automation and zero-manual-process initiatives across business domains.

ETL Developer → Senior ETL Developer

Earlier roles

Billing migration, retailer productivity, revenue reporting and B2B settlement initiatives using Ab Initio.

SELECTED PROJECTS

Ab Initio → Python / PySpark migration · Enterprise, ongoing (details under NDA)
Config-driven pipelines on a cloud data platform; parity-validated cut-over; 4,500+ generated runners.

legacy-etl-to-pyspark · Open source, 2026 · github.com/yuzansama/legacy-etl-to-pyspark
NDA-free demo of my migration approach: a legacy ETL job rebuilt in PySpark (explicit schema, null-safe filter with quarantine, unique-key lookup check, decimal money, idempotent partition writes) with a record-level parity check against legacy output; CI on PySpark 3.5 and 4.0.

ONE.Charity · Hackathon, 2026 · Solidity, Hardhat, OpenZeppelin, Next.js, ethers v6 · onecharity.vercel.app
Zero-fee donation platform on Ethereum (Sepolia). Milestone-based fund release after independent verification, conflict-of-interest guards, withdrawal delay and donor refunds; 38 automated contract tests including security regression tests.

Titik Asal Kopi · 2026 · Next.js 16, TypeScript, Tailwind v4, Google Apps Script
E-commerce site for a coffee brand: variant catalog, cart, WhatsApp checkout and order tracking backed by Google Sheets via a privacy-first API (field allow-list, enumeration-safe responses, formula-injection guard). Delivered with BRD, architecture and QA plan.

Loka Journey · Loka Global Pacific · 2026 · HTML/CSS/JS, PHP · lokaglobalpacific.com
Built the Loka Journey travel section: Asia tour packages and a Japan personal-shopper (jastip) catalog driven by CSV data, with admin panels and a lightweight PHP settings API.

EDUCATION

Bachelor of Electrical Engineering — Telkom University 2011 – 2016